Artificial Neural Networks and Deep Architectures, DD2437

Lab assignment 2: Radial basis functions, competitive learning and self-organisation

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1 Main objectives and scope of the assignment

Our major goals in the assignment were:

- To implement and evaluate Radial Basis Function (RBF) networks for regression, comparing analytical batch least-squares against the iterative online delta rule.
- To investigate the impact of network capacity, kernel width, and noise on generalisation, contrasting RBF performance with that of a standard Multi-Layer Perceptron (MLP).
- To apply Competitive Learning (CL) as a vector quantisation strategy for the automatic, data-driven positioning of RBF centres, thereby mitigating the dead-unit problem.
- To implement the core Self-Organising Map (SOM) algorithm entirely from scratch to perform unsupervised dimensionality reduction, topological ordering, and data clustering.
- To mathematically adapt the SOM neighbourhood function across three distinct topological structures: 1D linear (animal species), 1D circular (Travelling Salesman Problem), and 2D grid maps (parliamentary voting behaviour).

This assignment assumes fully observable, static datasets. To ensure a solid understanding of competitive learning dynamics, we implemented all core RBF and SOM algorithms from scratch in Python.

2 Methods

The algorithms were implemented in Python, heavily utilising the NumPy library for vectorised matrix operations, distance calculations, and linear algebra routines (e.g., solving the normal equations). Pandas was used to parse and clean the raw data files, while Matplotlib provided the visualisations. For the regression tasks in Part I, Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) were adopted to evaluate network approximation and generalisation. In Part II, the SOM Best Matching Unit (BMU) was determined via vectorised Euclidean distance, while the topological neighbourhood updates relied on topology-specific distance metrics: absolute difference (1D linear), minimum wrap-around distance (1D cyclic), and Manhattan distance (2D grid).

3 Results and discussion - Part I: RBF networks and Competitive Learning

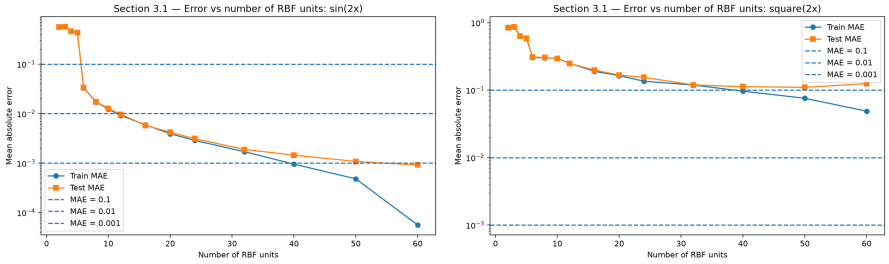
3.1 Batch mode training using least squares supervised learning of network weights

We evaluated Gaussian RBF networks on noise-free approximations of $\sin(2x)$ and $\text{square}(2x)$ using 63 samples per training/test set. The number of RBF units dictated approximation accuracy. For $\sin(2x)$, achieving training MAEs below 0.1, 0.01, and 0.001 required 6, 12, and 40 units, respectively, with the 40-unit model yielding a test MAE of 0.00144. The discontinuous $\text{square}(2x)$ function proved considerably harder to approximate, reaching only 0.096 training MAE with 40 units and failing to reach the 0.01 threshold even with 60 units. This difference is visible in Fig. 1.

Applying a sign function transformation to the continuous RBF output mapped the regression into a binary decision. This transformation achieved zero test error on the square wave using only 6 RBF units (compared to a raw MAE of 0.307), demonstrating its utility for classification or threshold-based decision systems where desired outputs are discrete states rather than continuous values.

3.2 Regression with noise

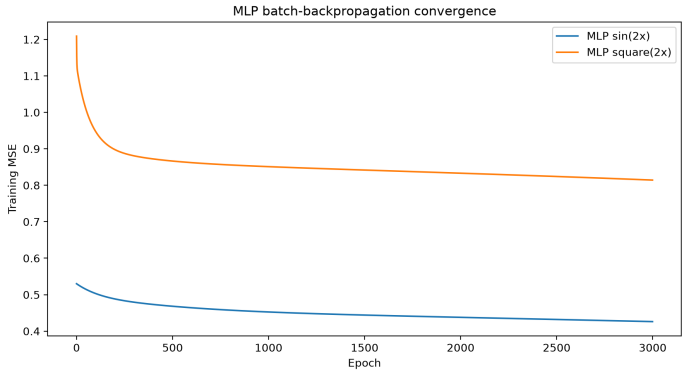
Adding zero-mean Gaussian noise (variance 0.1) shifted the evaluation focus to test MAE to measure generalisation. RBF width strongly controlled smoothing: extremely narrow kernels lacked overlap, while broad kernels oversmoothed. For $\sin(2x)$, an 8-unit network achieved a 0.266 test MAE with a width factor of 1, compared to 0.399 with a width of 0.25.



Figur 1: Training and test MAE as a function of the number of RBF units for $\sin(2x)$ and square$(2x)$ in the noise-free case.

For the online delta rule, learning rate (η) selection was critical. With 10 units, increasing η from 0.0001 to 0.01 improved test MAE from 0.518 to 0.253, but higher rates (0.05 and 0.1) slightly degraded generalisation (0.263 and 0.265 MAE, respectively). Centre positioning mattered most for smaller networks; for instance, manual vs. random placement on noisy $\sin(2x)$ yielded MAEs of 0.498 vs. 0.414 (5 units) and 0.261 vs. 0.265 (10 units). Testing these models on the original clean data yielded lower MAEs (0.084 for $\sin(2x)$; 0.209 for square$(2x)$) than on noisy data (0.254 and 0.306), indicating the models successfully learned the underlying functions.

Compared to a single-hidden-layer MLP trained via batch backpropagation with equal hidden units (15 for sine, 30 for square), the RBF networks achieved substantially better generalisation. The RBF noisy-test MAEs were 0.254 and 0.306, easily outperforming the MLP's respective errors of 0.563 and 0.811.



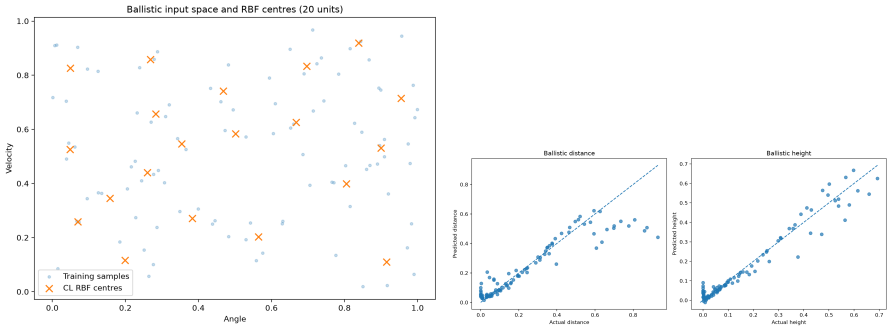
Figur 2: Training convergence of the single-hidden-layer MLP using batch backpropagation.

3.3 Competitive learning (CL) to initialise RBF units

We implemented competitive learning (CL) to automatically position RBF centres based on data distribution, using the delta rule for output weights. For a 10-

unit network on noisy $\sin(2x)$, CL achieved a test MAE of 0.253, matching the performance of manual, evenly-spaced positioning. To address the dead-unit problem observed in vanilla single-winner CL, a two-winner strategy was introduced. While both maintained 20 unique active centres, the two-winner approach slightly increased the quantisation error in this trial (0.128 vs. 0.086) while ensuring broader unit activation.

Finally, the CL-initialised RBF network successfully mapped a 2D ballistic dataset (angle and velocity to distance and height). Standardising the inputs prior to CL, a 20-unit configuration proved optimally compact, avoiding dead units entirely and achieving a test MAE of 0.044 and an RMSE of 0.079. Predicted points closely matched actual values, confirming the architecture's capability to learn multi-dimensional mappings despite noisy input data.



Figur 3: Ballistic experiment: (left) CL-initialised RBF centres in the angle–velocity input space and (right) predicted versus actual distance and height on the test set.

4 Results and discussion - Part II: Self-organising maps

4.1 Topological ordering of animal species

A 1D Self-Organising Map (SOM) was used to map 32 animal species, each characterised by 84 binary attributes, onto a linear sequence of 100 nodes. To preserve the linear topology, the neighbourhood was defined as the absolute 1D distance between node indices, shrinking linearly from a radius of 50 down to 1 over 20 epochs.

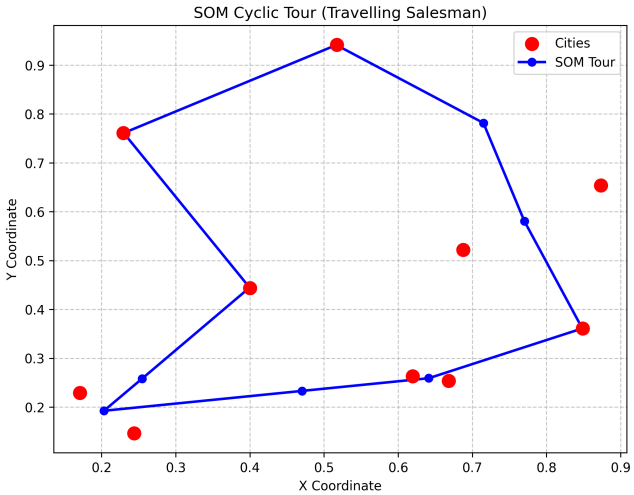
The resulting topological sort successfully grouped species based on their underlying biological and morphological similarities without any supervised labels. For example, insects (spider, beetle, mosquito) were clustered tightly at one end of the 100-node sequence, isolated from mammals. Within the mammalian group, further semantic sub-clustering emerged, such as felines (cat, lion) and large herbivores (elephant, giraffe, camel). This confirmed the SOM's ability to compress

an 84-dimensional feature space while preserving local topological relationships.

4.2 Cyclic tour

To address a variant of the Travelling Salesman Problem (TSP), a 1D SOM with 10 nodes was trained to find a minimal cyclic tour across 10 two-dimensional city coordinates. To enforce the cyclic nature of the tour, the topological distance metric was modified to account for ring wrap-around, calculating the minimum of the direct linear index distance and the circular complement. The neighbourhood radius was aggressively decayed from 2 to 0 to quickly force convergence.

As visualised in Figure 4, by extending the training to 100 epochs, the SOM weights successfully unfolded in the 2D space to form a clean, non-intersecting closed loop. In fact, individual weights converged to, or very closely approximated, the discrete city coordinates. The total path length was iteratively minimized, leading to a robust solution to the Travelling Salesman Problem.



Figur 4: Cyclic tour generated by a 10-node 1D circular SOM mapping to 10 city coordinates.

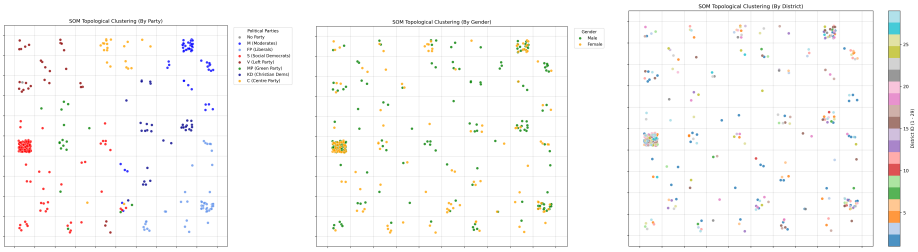
4.3 Clustering with SOM

Finally, a 2D SOM (10×10 grid) was trained to cluster 349 Swedish Members of Parliament (MPs) based exclusively on their voting records across 31 distinct parliamentary votes. The topological distance was calculated using the 2D Manhattan distance, with the neighbourhood radius decaying from 5 to 0 over 200 epochs. The number of epochs was increased with respect to the previous experiments, given the higher complexity of the problem. Jitter was applied

during plotting to separate multiple MPs mapping to the same Best Matching Unit (BMU).

As shown in Figure 5, without any prior knowledge of the MPs' political affiliations or demographics, the SOM mathematically discovered the traditional Swedish left-right political spectrum based strictly on their voting records. The left-wing bloc, predominantly the Social Democrats (red) and the Left Party (dark red), tightly clustered in the left regions of the grid, while the right-wing bloc, led by the Moderates (dark blue) and Liberals (light blue), mapped to the right regions. The physical distance across the SOM grid directly correlates with ideological distance in the 31-dimensional voting space.

Furthermore, projecting demographic and geographic labels onto the same trained topological map reveals the relative influence of these factors. As shown in Figure 5b, male and female MPs are completely mixed within every cluster. This mathematically proves that gender does not dictate voting behavior in the Swedish Parliament. A female Social Democrat votes identically to a male Social Democrat; they do not form a separate female voting bloc. Similarly, as we can see in Figure 5c, the colors representing the 29 geographic districts are scattered randomly across the grid. This proves that geography does not override party loyalty. An MP from District 1 votes with their party, not with other MPs from District 1 who belong to opposing parties.



(a) By Party (b) By Gender (c) By District

Figur 5: Topological clustering of 349 Swedish MPs on a 10×10 SOM grid, trained exclusively on 31 voting attributes. The uniform mixing in (b) and (c) contrasts with the strict ideological separation in (a).

5 Final remarks

The objective of this assignment was to organise and present complex data in an intuitive visual form understandable for humans.

In Part I, implementing RBF networks from scratch highlighted the importance of basis-function placement, width selection, and weight optimisation for function approximation. The experiments showed how RBFs can effectively model both smooth and discontinuous functions, while competitive learning provided a data-driven approach for positioning the centres. Comparing different learning

strategies, network sizes, and noisy conditions also demonstrated the trade-off between model complexity, generalisation, and robustness. In Part II, implementing the SOM algorithm from scratch reinforced our understanding of topological preservation. The transition from linear distance (animals) to circular distance (TSP) and finally to two-dimensional Manhattan distance (MP vote) demonstrated how changing the neighborhood function determines the structural development of the network. The visual discovery of the political spectrum from raw binary voting data was particularly interesting, demonstrating the SOM's ability to extract meaningful and human-interpretable patterns from high-dimensional datasets.